

1. Grey Level Transformation

(a) Purpose of grey level transformation in image enhancement

Grey level transformation (point transformation) maps each input pixel value r to an output value s using a function $s = T(r)$, applied independently and identically to every pixel regardless of its spatial location. Its main purposes in image enhancement are:

- Improve visual quality by enhancing contrast, brightness, or detail so the image is easier to interpret.
- Highlight specific intensity ranges of interest, such as dark, bright, or mid-tone regions (e.g. thresholding, grey-level slicing).
- Correct imaging defects such as under-exposure or over-exposure using log or power-law (gamma) transforms.
- Prepare images for further processing such as segmentation or edge detection, where certain features must be made to stand out.

Common types include the negative transformation, log transformation, power-law (gamma) transformation, piecewise-linear (contrast stretching), and grey-level slicing.

(b) Negative transformation

No specific pixel values were supplied with the question, so the sample set below is used to demonstrate the method.

Formula: $s = (L - 1) - r$, where $L = 256$ for an 8-bit image, so $s = 255 - r$.

Original (r)	Negative (s = 255 - r)
30	225
80	175
150	105
200	55
255	0

Dark pixels (near 0) become bright and bright pixels (near 255) become dark — the image appears like a photographic negative.